

Euclid's Elements — Book XI

Proposition 15

Formal Reconstruction — Technical Documentation

Jurek Róžański
Military University of Technology, Warsaw, Poland
jerzy.rozanski@wat.edu.pl

CGJTEAMLAB

If two straight lines meeting one another be parallel to two straight lines meeting one another, not being in the same plane, the planes through them are parallel.

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1 Introduction

Proposition XI.15 is a structural culmination of the parallelism and perpendicularity block developed in XI.8–XI.14. Euclid starts with two intersecting lines in one plane and two corresponding intersecting lines in another plane. If the corresponding pairs are parallel, then the two planes themselves are parallel.

The classical proof is remarkably economical. From the intersection point of the first pair, Euclid drops a perpendicular to the second plane by XI.11. Through its foot he draws two lines parallel to the two reference lines in that plane. Definition XI.3 makes the new normal perpendicular to both of these lines; XI.9 and I.29 transfer the two right angles back to the original pair. XI.4 then says that the same line is perpendicular to the first plane, and XI.14 finishes the proof.

The formal reconstruction preserves this outer architecture, but it exposes three hidden issues.

First, XI.11 requires the point B to lie outside the second plane ρ . Euclid’s diagram makes this look automatic, but Lean proves it from the two spatial parallelisms and the distinctness of the two planes.

Second, Euclid invokes I.31 twice through the foot G . The formal parallel-through-a-point construction requires the point to be outside the given line. XI.15 does not guarantee that G lies outside either reference line. The production proof therefore treats the cases $G \in m_i$ and $G \notin m_i$ uniformly by constructing a line q_i which is either literally m_i or parallel to m_i .

Third, the classical I.29 step is not used as one opaque angle theorem. The formal proof factors the transport of perpendicularity through the mechanism already developed in XI.8: neutral I.27 gives disjointness of two perpendiculars to the same transversal, and Hilbert Group IV identifies the unique parallel through the required point.

The result is a proof whose top-level DAG is very close to Euclid’s, while its local geometry is more explicit about plane carriers, exceptional incidence, and the exact place where Euclidean

parallel uniqueness enters.

2 Classical Configuration

Let AB and BC be two straight lines meeting at B , and let DE and EF be two straight lines meeting at E . Suppose

$$AB \parallel DE, \quad BC \parallel EF,$$

and suppose that the four lines are not all contained in one plane.

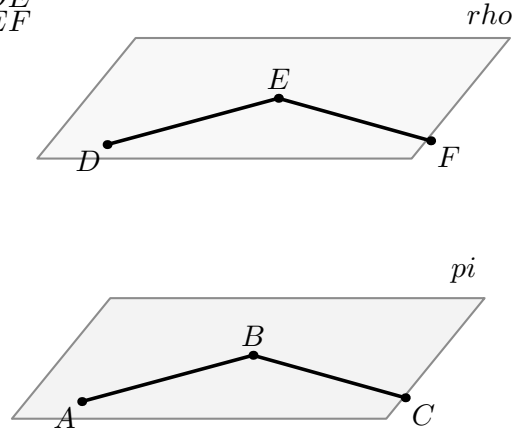
Write

$$\pi = \text{plane}(AB, BC), \quad \rho = \text{plane}(DE, EF).$$

The conclusion is

$$\pi \parallel \rho.$$

$$\begin{array}{l} AB \text{ parallel to } DE \\ BC \text{ parallel to } EF \end{array}$$



two intersecting pairs in distinct plane slices

Figure 1: The classical spatial configuration of Proposition XI.15. The lines AB, BC meet at B in plane π , while DE, EF meet at E in plane ρ . Corresponding carriers are parallel: $AB \parallel DE$ and $BC \parallel EF$. Geometry internal to a displayed plane is black; the separation between the two plane slices is spatial rather than an additional geometric connector.

Diagrammatic audit. Three stable geometric states are sufficient for XI.15. Figure 1 records the classical input. Figure 2 records the XI.11 normal and the two formal replacements for Euclid’s I.31 constructions, including the hidden exceptional branch $G \in m_i$. Figure 3 records the plane-slice mechanism which transports perpendicularity across a parallel pair.

No fourth figure is introduced for the final state

$$n \perp \pi, \quad n \perp \rho,$$

because that state is exactly the input configuration of Proposition XI.14 and would duplicate the already documented XI.14 geometry.

3 Statement

Euclid’s statement is:

If two straight lines meeting one another be parallel to two straight lines meeting one another, not being in the same plane, the planes through them are parallel.

The source-style mathematical form is therefore:

Theorem 1 (Euclid XI.15). *Let l_1, l_2 be distinct lines meeting in a plane π , and let m_1, m_2 be distinct lines meeting in a plane ρ . Suppose*

$$l_1 \parallel m_1, \quad l_2 \parallel m_2,$$

and suppose the four lines do not lie in one common plane. Then

$$\pi \parallel \rho.$$

The production file exposes a slightly stronger public theorem. It requires the first pair l_1, l_2 to meet at B , but it does not require an explicit second intersection point E or the inequality $m_1 \neq m_2$. Those extra data are not needed by the proof once the second pair is known to lie in ρ and the two corresponding spatial parallelisms are given.

The current public declaration is:

```

theorem euclid_proposition_11_15
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (pi rho : S.Plane)
  (l1 l2 : PlaneLine Geo pi)
  (m1 m2 : PlaneLine Geo rho)
  (B : PlanePoint Geo pi)
  (hB11 : H.OnLine B.1 l1.1)
  (hB12 : H.OnLine B.1 l2.1)
  (hl12 : Ne l1 l2)
  (hParallel1 :
    HilbertSpaceLinesParallel Geo l1.1 m1.1)
  (hParallel2 :
    HilbertSpaceLinesParallel Geo l2.1 m2.1)
  (hNoCommonPlane :
    Not (exists omega : S.Plane,
      HilbertLineInPlane Geo l1.1 omega /\
      HilbertLineInPlane Geo l2.1 omega /\
      HilbertLineInPlane Geo m1.1 omega /\
      HilbertLineInPlane Geo m2.1 omega)) :
    HilbertSpacePlanesParallel Geo pi rho

```

For source comparison the file also provides

```
euclid_proposition_11_15_intersecting_pairs
```

which adds an explicit point $E \in m_1 \cap m_2$ and the hypothesis $m_1 \neq m_2$. Its proof is a direct projection to the stronger public theorem.

4 Classical Proof

Starting from B , Euclid applies XI.11 to draw

$$BG \perp \rho,$$

where $G \in \rho$.

Through G , Proposition I.31 gives two lines in ρ :

$$GH \parallel ED, \quad GK \parallel EF.$$

Since $BG \perp \rho$, Definition XI.3 gives

$$BG \perp GH, \quad BG \perp GK.$$

Now

$$BA \parallel ED \quad \text{and} \quad GH \parallel ED.$$

By XI.9,

$$BA \parallel GH.$$

Similarly,

$$BC \parallel GK.$$

Proposition I.29 transfers the two right angles:

$$BG \perp BA, \quad BG \perp BC.$$

The two lines BA and BC meet at B , so XI.4 yields

$$BG \perp \pi.$$

But by construction

$$BG \perp \rho.$$

Therefore XI.14 applies to the same straight line BG and gives

$$\pi \parallel \rho.$$

The classical local dependency pattern is

$$\begin{array}{c}
 AB \parallel DE, \quad BC \parallel EF \\
 \downarrow \\
 \text{XI.11 : } BG \perp \rho \\
 \downarrow \\
 \text{I.31 : } GH \parallel ED, \quad GK \parallel EF \\
 \downarrow \\
 \text{XI.Def.3 : } BG \perp GH, \quad BG \perp GK \\
 \downarrow \\
 \text{XI.9 + I.29} \\
 \downarrow \\
 BG \perp BA, \quad BG \perp BC \\
 \downarrow \\
 \text{XI.4} \\
 \downarrow \\
 BG \perp \pi \\
 \downarrow \\
 \text{XI.14} \\
 \downarrow \\
 \pi \parallel \rho.
 \end{array}$$

Thus XI.15 is classically a short synthesis theorem: XI.11 constructs the normal, XI.4 upgrades two line-line perpendicularities to a line-plane perpendicularity, and XI.14 converts a common normal into plane parallelism.

5 The Hidden I.31 Exceptional Case

The most important local correction in the formal reconstruction occurs before the classical I.31 step.

Euclid says simply: through G , draw a line parallel to ED and a line parallel to EF . In a formal Hilbert construction, however, a parallel through a point is constructed only when the point lies outside the given line. XI.15 does not imply

$$G \notin DE \quad \text{or} \quad G \notin EF.$$

The production theorem therefore uses the proposition-local helper

```
hilbert_XI15_parallel_or_equal_through_point_in_plane
```

with the following mathematical content. Given a line $m \subset \rho$ and a point $G \in \rho$, it returns a line $q \subset \rho$ through G such that

$$q = m \quad \text{or} \quad m \parallel q.$$

There are two branches:

$$\begin{aligned} G \in m &\implies q := m, \\ G \notin m &\implies \text{use the XI.12 parallel-through-point construction.} \end{aligned}$$

In the second branch, the XI.12 helper returns spatial parallelism together with a carrier plane. XI.15 then proves that the constructed line actually lies in the intended plane ρ , using uniqueness of the plane through a line and an external point.

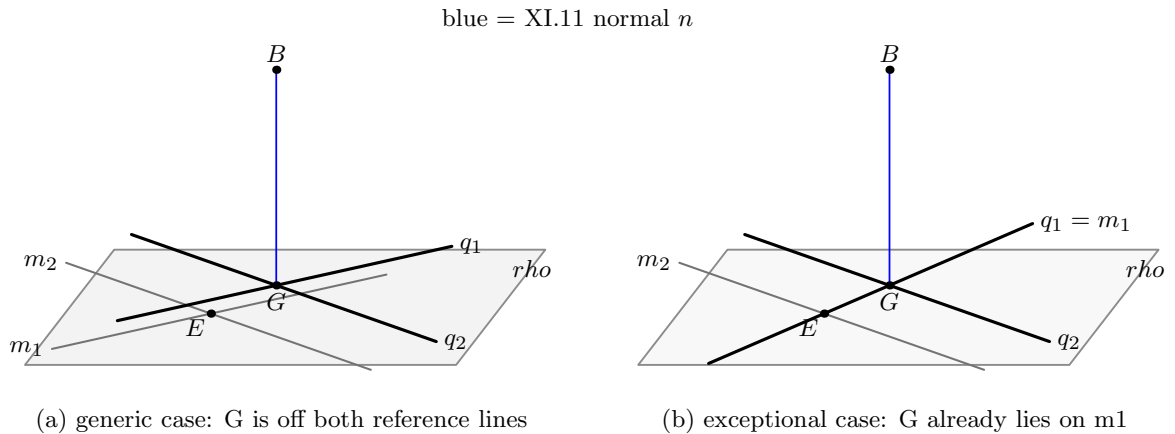


Figure 2: The formal replacement for Euclid's two I.31 instructions. In the generic branch, lines $q_1, q_2 \subset \rho$ through G are parallel to the reference lines m_1, m_2 . In the exceptional branch $G \in m_1$, the correct choice is simply $q_1 = m_1$; only the second reference line requires a new parallel. The blue line is the genuinely spatial normal n , while all black and gray lines lie in ρ .

This case split is not a Lean artifact. It is the precise formal content of the geometric fact that the instruction “draw a parallel through G ” degenerates to “use the given line itself” when G already lies on the carrier.

6 What is genuinely three-dimensional here?

XI.15 alternates repeatedly between ambient space and induced plane geometry.

The genuinely spatial data are:

- the two reference planes π, ρ ;
- the two pairs of corresponding spatially parallel lines;
- the condition that all four lines do not lie in one ambient plane;
- the point B , which must be proved to lie outside ρ ;
- the XI.11 normal n from B to ρ ;
- the final line-plane perpendicularities $n \perp \pi$ and $n \perp \rho$;
- the spatial target $\text{HilbertSpacePlanesParallel}(\pi, \rho)$.

There are also auxiliary carrier planes which are genuinely spatial. Every assertion of

$$\text{HilbertSpaceLinesParallel}(l, q)$$

contains a plane witnessing that l and q are coplanar. The perpendicularity-transport lemma uses precisely this carrier plane, denoted σ , to move from ambient line data to an induced $\text{PlaneGeo}(\sigma)$.

The planar work occurs in two distinct kinds of slices:

1. the reference plane ρ , where the lines through the foot G are constructed;
2. a carrier plane σ for each parallel pair $l \parallel q$, where the I.27/Group IV perpendicularity transport is performed.

Thus XI.15 is not one planar proof placed in space. The proof changes planes for mathematically different reasons: first to construct reference directions inside ρ , then to compare a transversal with a particular parallel pair inside its own carrier plane.

7 Formal Statement and Normalized Core

The substantial theorem is first isolated under explicit plane distinctness:

```
theorem euclid_proposition_11_15_normalized
...
(pi rho : S.Plane)
(l1 l2 : PlaneLine Geo pi)
(m1 m2 : PlaneLine Geo rho)
```

```

(B : PlanePoint Geo pi)
...
(hl12 : Ne l1 l2)
(hParallel1 :
  HilbertSpaceLinesParallel Geo l1.1 m1.1)
(hParallel2 :
  HilbertSpaceLinesParallel Geo l2.1 m2.1)
(hPlanesNe : Ne pi rho) :
HilbertSpacePlanesParallel Geo pi rho

```

The public theorem replaces the explicit hypothesis

$$\pi \neq \rho$$

by Euclid's source condition that the four lines are not contained in one common plane. If $\pi = \rho$, then all four displayed lines lie in that same plane, contradicting the public hypothesis. The public wrapper is therefore mathematically thin.

This separation is useful. The normalized theorem expresses the actual geometric engine. The public theorem expresses the source-facing configuration.

8 Spatial / Planar Architecture Used

The ambient layer is:

```

HilbertIncidence
HilbertPlaneIncidence
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean

```

The principal spatial relations are:

```

HilbertLineInPlane
HilbertSpaceLinesParallel
HilbertLinesDisjoint
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
HilbertSpacePlanesParallel

```

The induced plane layer uses:

```

PlanePoint
PlaneLine
PlaneOnLine
PlaneGeo

```

and the standard ambient/plane bridges, in particular the bridge for line-line perpendicularity.

A permanent Book XI instance rule is respected throughout: the proof does not install global ambient instances

```
HilbertOrder Geo
HilbertCongruence Geo
```

for three-space. Order and planar congruence are supplied through explicit PlaneGeo slices; spatial congruence remains in HilbertSpaceCongruence.

9 Proof Architecture of the Reconstruction

9.1 Proving $B \notin \rho$

The normalized theorem cannot invoke XI.11 until it has proved that the intersection point B of l_1, l_2 lies outside the second plane.

The helper

```
hilbert_XI15_intersection_point_off_other_plane
```

is purely incidence-theoretic.

Assume $B \in \rho$. The parallelism $l_1 \parallel m_1$ provides a carrier plane σ_1 containing both lines. Since the two parallel lines are disjoint, $B \notin m_1$. Both σ_1 and ρ therefore contain the same line m_1 and the same external point B . Plane uniqueness gives

$$\sigma_1 = \rho,$$

hence

$$l_1 \subset \rho.$$

The same argument gives

$$l_2 \subset \rho.$$

But the distinct intersecting lines l_1, l_2 determine a unique plane. They already lie in π , and now they also lie in ρ . Therefore

$$\pi = \rho,$$

contradicting the normalized hypothesis.

The file also contains the symmetric package

```
hilbert_XI15_intersection_points_off_opposite_planes
```

which proves both $B \notin \rho$ and the analogous source-style statement $E \notin \pi$. The main normalized proof needs only the first half.

9.2 XI.11 constructs the normal

With $B \notin \rho$ established, the production proof invokes

```
euclid_proposition_11_11
```

and obtains a line n and a foot G satisfying

$$B \in n, \quad n \perp \rho \text{ at } G.$$

This is packaged by

```
hilbert_XI15_normal_from_intersection_to_other_plane
```

and then combined with the two reference-line constructions in

```
hilbert_XI15_normal_and_reference_lines
```

to produce

$$q_1, q_2 \subset \rho, \quad G \in q_1 \cap q_2,$$

with

$$q_i = m_i \quad \text{or} \quad m_i \parallel q_i.$$

9.3 Recovering $l_i \parallel q_i$

The original hypotheses give

$$l_i \parallel m_i.$$

If $q_i = m_i$, there is nothing to prove.

In the genuine parallel branch

$$m_i \parallel q_i,$$

the proof needs strict transitivity of spatial parallelism. This is slightly subtle because project parallelism is irreflexive: a line is not parallel to itself.

The helper

```
hilbert_XI15_spaceParallel_transitive_distinct
```

therefore assumes explicitly that the outer lines are distinct.

Its proof again separates two spatial configurations:

- if l, m, q lie in one plane, Hilbert Group IV gives uniqueness of the parallel to m through a point;
- if the three lines do not lie in one common plane, XI.9 supplies the required spatial transitivity.

In XI.15 the required distinctness $l_i \neq q_i$ follows from $B \notin \rho$. Indeed, if $l_i = q_i$, then $B \in q_i \subset \rho$.

Thus the proof obtains

$$l_1 \parallel q_1, \quad l_2 \parallel q_2.$$

9.4 Definition XI.3 gives the two local perpendicularities

Since

$$n \perp \rho \text{ at } G$$

and

$$q_i \subset \rho, \quad G \in q_i,$$

Definition XI.3 is represented directly by `HilbertLinePerpendicularPlaneAt.perpendicular_to_line`. It gives

$$n \perp q_1 \text{ at } G, \quad n \perp q_2 \text{ at } G.$$

The remaining problem is to transport these perpendicularities from q_i to the parallel lines l_i .

9.5 The formal I.29 replacement

The central transport theorem is

```
hilbert_XI15_perpendicular_transfer_across_parallel
```

and its line-plane wrapper is

```
hilbert_XI15_plane_normal_perpendicular_to_parallel_line
```

Suppose

$$l \parallel q,$$

the transversal n meets q at G and l at B , and

$$q \perp n \text{ at } G.$$

The spatial parallelism $l \parallel q$ supplies a common carrier plane σ . Since $B, G \in n$ and $B \neq G$, the whole transversal n also lies in σ .

The proof now passes to

$$\text{PlaneGeo}(\sigma).$$

Through B , it constructs a line p perpendicular to n . Hence

$$p \perp n \text{ at } B, \quad q \perp n \text{ at } G.$$

The feet are distinct. The neutral XI.8 helper representing the I.27 mechanism therefore gives

$$p \cap q = \emptyset.$$

But l is also disjoint from q , and both l and p pass through B . Hilbert Group IV gives uniqueness of the parallel through B :

$$p = l.$$

Therefore

$$l \perp n \text{ at } B,$$

and by symmetry

$$n \perp l \text{ at } B.$$

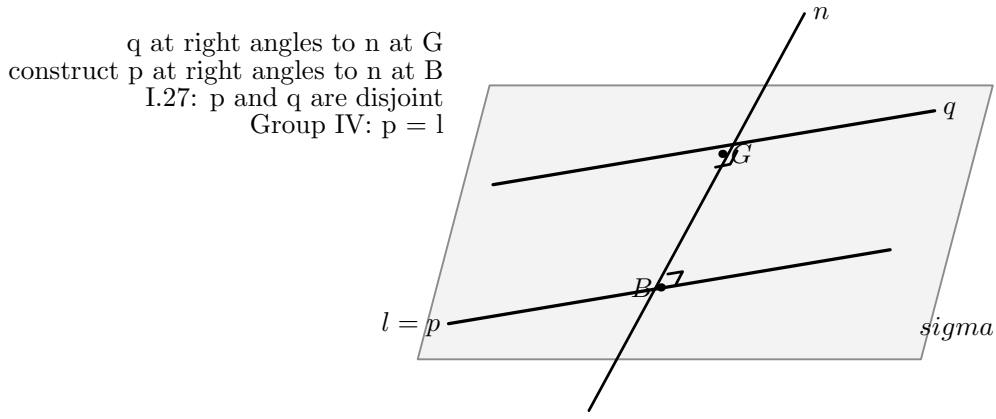


Figure 3: The formal replacement for Euclid’s I.29 step, carried out in the common plane σ of $l \parallel q$. The line q is perpendicular to the transversal n at G . A perpendicular p to n is erected at B . Neutral I.27 gives $p \cap q = \emptyset$; Group IV then identifies p with the already given parallel l . The final lower carrier is therefore labelled $l = p$.

This is exactly the logical role of Euclid’s I.29, but factored according to the current Hilbert architecture:

$$\text{I.27} + \text{Group IV uniqueness} \implies \text{right-angle transport across parallels.}$$

9.6 XI.4 makes the normal perpendicular to π

Applying the preceding transport twice gives

$$n \perp l_1 \text{ at } B, \quad n \perp l_2 \text{ at } B.$$

The lines l_1, l_2 are distinct and meet at B , so the production theorem

euclid_proposition_11_4

gives

$$n \perp \pi \text{ at } B.$$

This is one of the cleanest stages of the formal proof: the substantial work has already been normalized into exactly the XI.4 interface.

9.7 XI.14 closes the proof

From XI.11 we still have

$$n \perp \rho \text{ at } G.$$

From XI.4 we have

$$n \perp \pi \text{ at } B.$$

The normalized theorem assumes $\pi \neq \rho$, so XI.14 applies directly:

$$\pi \parallel \rho.$$

The final line of the normalized proof is therefore an actual invocation of

```
euclid_proposition_11_14
```

rather than a new plane-separation argument.

9.8 The source-facing wrapper

The public theorem assumes that the four displayed lines do not all lie in one plane. It first derives

$$\pi \neq \rho.$$

If the two planes were equal, then l_1, l_2, m_1, m_2 would all lie in π , contradicting the source hypothesis.

The public theorem then calls the normalized core.

The separate theorem

```
euclid_proposition_11_15_intersecting_pairs
```

adds the explicit second intersection point and line distinctness appearing in Euclid's wording. Since those data are not needed internally, this source-style theorem is intentionally only a wrapper.

10 Lean Formalization

The production file imports:

```
import CGJteamLab.Proposition11_14
import CGJteamLab.Proposition11_12
import CGJteamLab.Proposition11_9
import CGJteamLab.Proposition11_8
import CGJteamLab.Proposition11_4
```

Mathematically, the proof also calls XI.11 through the imported Book XI dependency chain.

The proposition-local helpers fall into four conceptual groups.

Spatial incidence normalization.

```
hilbert_XI15_spaceLinesParallel_symm
hilbert_XI15_intersection_point_off_other_plane
hilbert_XI15_intersection_points_off_opposite_planes
```

The XI.11/I.31 construction package.

```
hilbert_XI15_parallel_or_equal_through_point_in_plane
hilbert_XI15_normal_from_intersection_to_other_plane
hilbert_XI15_normal_and_reference_lines
```

Parallel and perpendicular transport.

```
hilbert_XI15_perpendicular_transfer_across_parallel  
hilbert_XI15_plane_normal_perpendicular_to_parallel_line  
hilbert_XI15_spaceParallel_transitive_distinct
```

Final theorem layer.

```
euclid_proposition_11_15_normalized  
euclid_proposition_11_15  
euclid_proposition_11_15_intersecting_pairs
```

This grouping reflects stable mathematical responsibilities, not the chronology of the test files used during development.

11 Hidden Formal Data

11.1 The XI.11 starting point must be outside ρ

Euclid begins XI.11 from B without a separate argument that $B \notin \rho$. The production proof derives this by a nontrivial spatial incidence argument. This is the first hidden obligation in XI.15.

11.2 The I.31 point may already lie on the reference line

The foot G may lie on m_1 , on m_2 , or on neither. The formal construction cannot assume the generic diagrammatic case. The equal-or-parallel helper packages all possibilities without changing the outer proof.

11.3 Spatial parallelism carries a plane witness

The predicate `HilbertSpaceLinesParallel` is not merely a disjointness relation. It also stores a common carrier plane. XI.15 uses that carrier plane to create the `PlaneGeo` slice in which perpendicularity transport is proved.

11.4 Spatial parallelism is strict

Because parallel lines are disjoint, the project relation is irreflexive. A transitivity lemma therefore needs the extra condition that the two outer lines are distinct. In XI.15 that condition is not assumed separately; it is derived from $B \notin \rho$.

11.5 The feet B and G must be distinct

The perpendicularity-transport theorem requires two distinct feet on the transversal n . If $B = G$, then B would lie on both parallel lines l and q , contradicting their disjointness.

11.6 The second source intersection is redundant

Euclid describes both pairs as intersecting. The public production theorem needs only the first explicit intersection $B \in l_1 \cap l_2$. The second intersection is mathematically compatible with the theorem but is not used by the proof.

This is a genuine strengthening, not an omission from the formal configuration. The source-style corollary records the literal Euclidean data separately.

12 Relationship with Earlier Book XI Propositions

XI.15 is unusually transparent as a consumer of the preceding Book XI infrastructure.

XI.4. This proposition is used exactly at the classical place. Once the formal proof has obtained

$$n \perp l_1, \quad n \perp l_2$$

at their common point B , XI.4 upgrades the pair to

$$n \perp \pi.$$

XI.8. The final theorem XI.8 is not the target of the proof, but XI.15 reuses its neutral helper for two coplanar lines perpendicular to the same transversal at distinct points. This is the formal I.27 engine inside the replacement for I.29.

XI.9. XI.9 appears in two roles. It is part of the classical proof of $BA \parallel GH$ and $BC \parallel GK$. Formally, it is used inside the strict transitivity helper when three spatial parallel lines do not have one common carrier plane.

XI.11. XI.11 supplies the normal from B to the second plane. Before it can be called, XI.15 proves the hidden hypothesis $B \notin \rho$.

XI.12. The plane-local parallel-through-point helper developed in XI.12 is reused in the generic branch $G \notin m_i$. XI.15 is therefore the first later proposition where that helper becomes genuine reusable infrastructure rather than merely proposition-local convenience.

XI.14. XI.14 is the terminal theorem. Once the same line n is known to be perpendicular to both π and ρ , the conclusion of XI.15 is immediate.

XI.10 is conceptually related because it starts from the same kind of configuration—two pairs of corresponding parallel lines in distinct spatial plane slices. But XI.15 does not need XI.10’s directed segment parallelism. Orientation of selected rays matters for equality of angles in XI.10; it is irrelevant to plane parallelism in XI.15.

13 Relationship with Book I / Book Zero / Hilbert Infrastructure

The classical Book I dependencies are I.31 and I.29.

The formal reconstruction preserves the role of I.31 but makes its precondition explicit. The generic branch uses the existing parallel-through-point construction in a plane slice; the exceptional branch uses the original line itself.

The classical I.29 step is represented more structurally. The current proof uses:

- a neutral I.27-type theorem to show that two lines perpendicular to the same transversal at distinct points are disjoint;
- Hilbert Group IV to identify the unique parallel through the required point.

Thus XI.15 genuinely uses Euclidean parallel geometry. This is visible already in the production signature:

```
[HSE : HilbertSpaceEuclidean Geo]
```

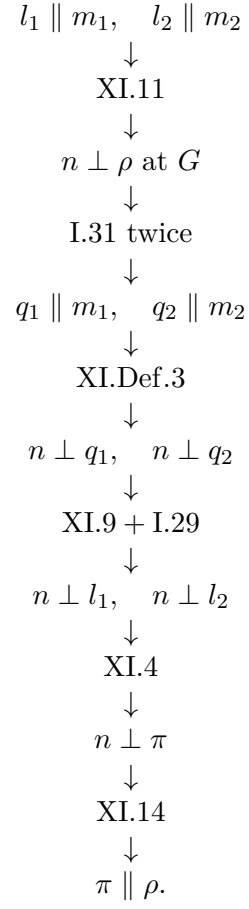
The use of Group IV is mathematically substantive, not merely inherited module noise.

No continuity principle is required. No coordinate system, vector, scalar product, numerical angle measure, or trigonometric function occurs in the reconstruction.

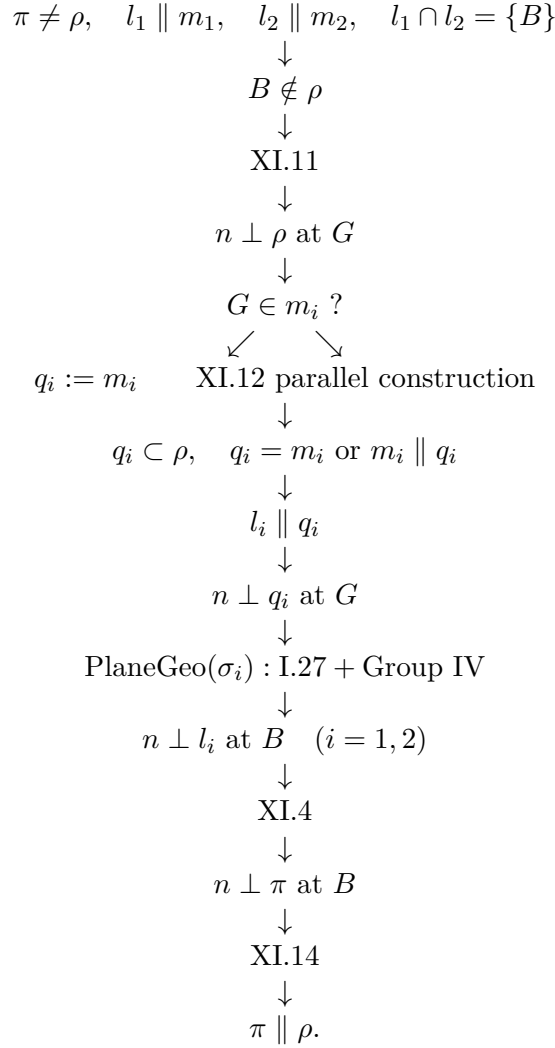
Book Zero contributes only the already established synthetic infrastructure used transitively by the right-angle and planar construction layers. XI.15 introduces no new proposition-specific geometric axiom.

14 Dependency Summary

The classical DAG is:



The formal DAG is:



The formal proof is longer around the two classical line constructions and the I.29 step, because the diagram suppresses exceptional incidence and carrier-plane information. It is shorter at the public boundary: the source-style second intersection point E is not required by the geometric core.

15 Audit Status

The documentation was checked against the current production `Proposition11_15.lean`. The user reported that the production file, the separate axiom-audit file, and the full project build all compile cleanly.

The separate audit reports the standard accepted project baseline and no new proposition-specific geometric axiom. The mathematically important foundation fact is the one visible in the theorem signature itself: XI.15 genuinely uses the Euclidean spatial layer `HilbertSpaceEuclidean`, while no continuity assumption enters.

The figures follow the established Book XI convention: displayed plane geometry is black or gray, genuinely spatial lines are blue, hidden spatial segments are blue dashed when needed, and plane fills remain light.

16 Conclusion

Proposition XI.15 is a particularly successful example of the Book XI architecture reaching maturity.

Euclid’s top-level idea survives almost unchanged. Construct a normal to the second plane, transfer its right angles to the two lines of the first plane, use XI.4 to make it a normal to that plane as well, and conclude by XI.14 that the planes are parallel.

What formalization adds is not a different geometry but a precise account of the information carried by the classical diagram. It proves that the first intersection point is genuinely outside the second plane; it repairs the hidden exceptional case in the two I.31 constructions; it records the carrier planes implicit in spatial parallelism; and it factors the I.29 transport into neutral perpendicular geometry plus Group IV uniqueness.

The resulting proof also clarifies the relation between source statement and formal theorem. Euclid presents two intersecting pairs, but the second explicit intersection is not needed by the geometric core. The production theorem therefore proves a slightly stronger statement, while a thin source-style corollary retains the literal Euclidean configuration.

At the level of the Book XI DAG, XI.15 is an important closure theorem: XI.11 supplies the common normal, XI.12 supplies the reusable parallel-through-point construction, XI.8 and XI.9 supply the parallel transport machinery, XI.4 upgrades two right angles to a line-plane perpendicularity, and XI.14 converts the common normal into the final parallelism of planes. No coordinates or metric algebra are needed; the proof remains entirely synthetic.